

Algorithmic Trading

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Case Study 01: Post Trade Analysis

Assignment:

Perform post-trade analysis and develop a Broker Report Card.

File: CaseStudy_PostTrade.csv

Perform analysis for the following categories:

- i) All Orders
- ~~ii) Small Orders (Size $\leq$ 5% ADV)~~
- ~~iii) Large Orders (Size $>$ 5% ADV)~~

Calculate Costs and Measure Trade Performance for all orders and by broker. This includes:

- Implementation Shortfall
 - Total IS, Delay, Execution, Opportunity, Fixed (use \$0.01 commission)
- Arrival Cost
- VWAP Slippage
- Value-Add
- Z-Score

Calculate data in the “Broker Report Card” summary table below.

Analyze the Reports based on the Broker Report Card

- 1) What brokers provided the best performance
- 2) What brokers underperformed expectations
- ~~3) Did any brokers perform better or worse for any category or trades (e.g., outperformed expectations for large orders and/or underperformed expectations for small orders)?~~
- ~~4) Which broker(s) would you recommend for the different trade categories~~

To Upload to BB:

- Word file or PDF with your replies to questions 1-4 above.
- Your Broker Report Card Table with your calculated results
- Your Python Notebook

Broker Report Card

All Orders												
			Implementation	Delay	Execution	Opportunity	Fixed	Arrival	VWAP			
Broker	# Orders	Avg Size	Shortfall (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Cost (bp)	Slippage (bp)	Value-Add (bp)	Z-Score	
All Brokers												
Broker A												
Broker B												
Broker C												
Broker D												
Broker E												
Small Orders (Size <= 5% ADV)												
			Implementation	Delay	Execution	Opportunity	Fixed	Arrival	VWAP			
Broker	# Orders	Avg Size	Shortfall (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Cost (bp)	Slippage (bp)	Value-Add (bp)	Z-Score	
All Brokers												
Broker A												
Broker B												
Broker C												
Broker D												
Broker E												
Large Orders (Size > 5% ADV)												
			Implementation	Delay	Execution	Opportunity	Fixed	Arrival	VWAP			
Broker	# Orders	Avg Size	Shortfall (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Cost (\$)	Cost (bp)	Slippage (bp)	Value-Add (bp)	Z-Score	
All Brokers												
Broker A												
Broker B												
Broker C												
Broker D												
Broker E												